# ALGORITHM & FLOWCHART

# Agenda

- Pseudocode
- Algorithm
- Flowchart
- Examples
- References

# Algorithm & Flowchart for Problem Solving

**PROBLEM SOLVING**

- First producea general algorithm (one can use *pseudocode*)
- Refine the algorithm successively to get step by step detailed *algorithm* that is very close to a computer language.
- Draw flowcharts which show how data flows from source documents through the computer to final distribution to users
- Implement the solution by developing a program using any high level programming language such as C, C++, Java, Python

# Pseudocode

| Pseudocode | C language code |
|---|---|
| LOOP { | while(1) { |
| EXIT LOOP | break; |
| IF (conditions) { | if (conditions) { |
| ELSE IF (conditions) { | else if (conditions) { |
| ELSE { | else |
| INPUT a | scanf(“%d”,&a); |
| OUTPUT “Value of a:” a | printf(“Value of a: %d”,a); |
| + - * / % | + - * / % |
| = | == |
| <- | = |
| != | != |
| AND | && |
| OR | \|\| |
| NOT | ! |

# Algorithm

- What is an algorithm?
  - ✔ Algorithms are self contained set of operations to be performed step by step.
  - ✔ A process or set of rules to be followed in calculations or other problem-solving operations, especially by a computer.

# Algorithm

<u>Example:</u> Write an algorithm to find the sum of 2 numbers.

<u>Algorithm</u>

Step 1: Start

Step 2: Read A, B

Step 3: S= A + B

Step 4: Display Sum 'S'

Step 5: Stop

# Algorithm

1. Write an algorithm to find the remainder after division of 2 numbers.
2. Write an algorithm to convert temperature from Fahrenheit to Celsius.

# Properties of an algorithm

- Finiteness
- Definiteness
- Generality
- Effectiveness
- Input-Output

# Flowchart

- What is Flow Chart ?
  - ✔ Pictorial/ graphical representation of an algorithm is known as flow chart.
  - ✔ It uses some special symbol to represent instruction set.
  - ✔ Information system flowcharts show how data flows from source documents through the computer to final distribution to users.

# Flowchart Symbols

| Function | Symbol |
| --- | --- |
| Start/End |  |
| Processing |  |
| Input/output |  |
| Decision |  |
| Connector |  |
| Flow lines |  |

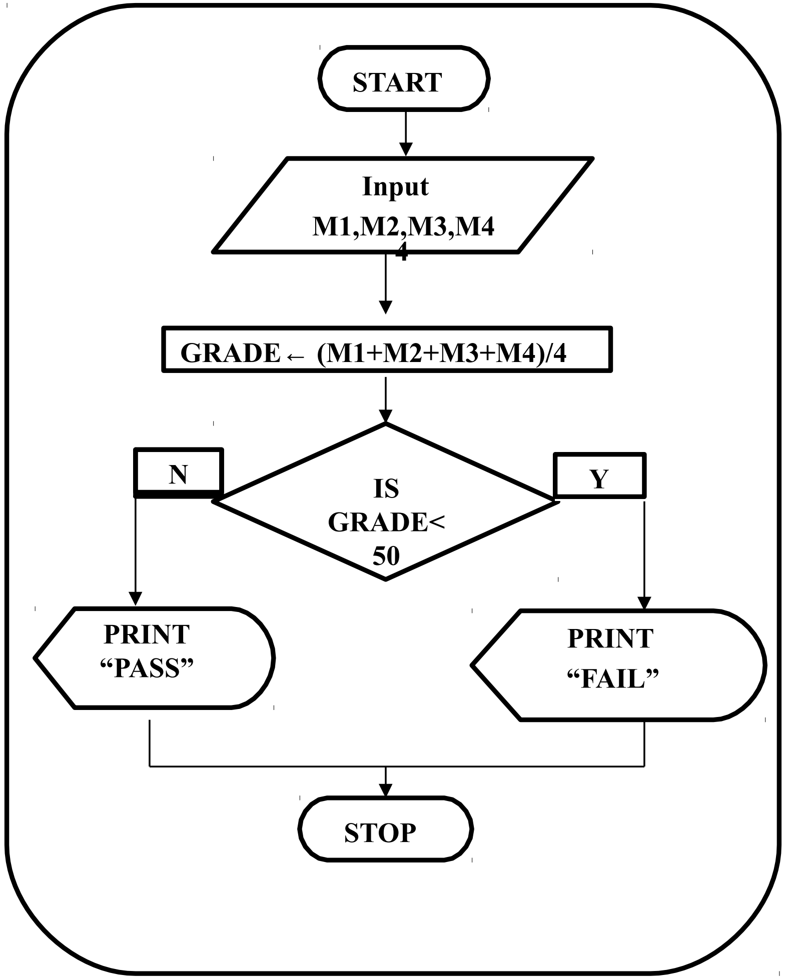
# PSEUDOCODE, ALGORITHM & FLOWCHART

Example: Write an algorithm to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

Pseudocode:

```
Input a set of 4 marks
Calculate their average by summing and dividing by 4
if average is below 50
      Print "FAIL"
else
      Print "PASS"
```

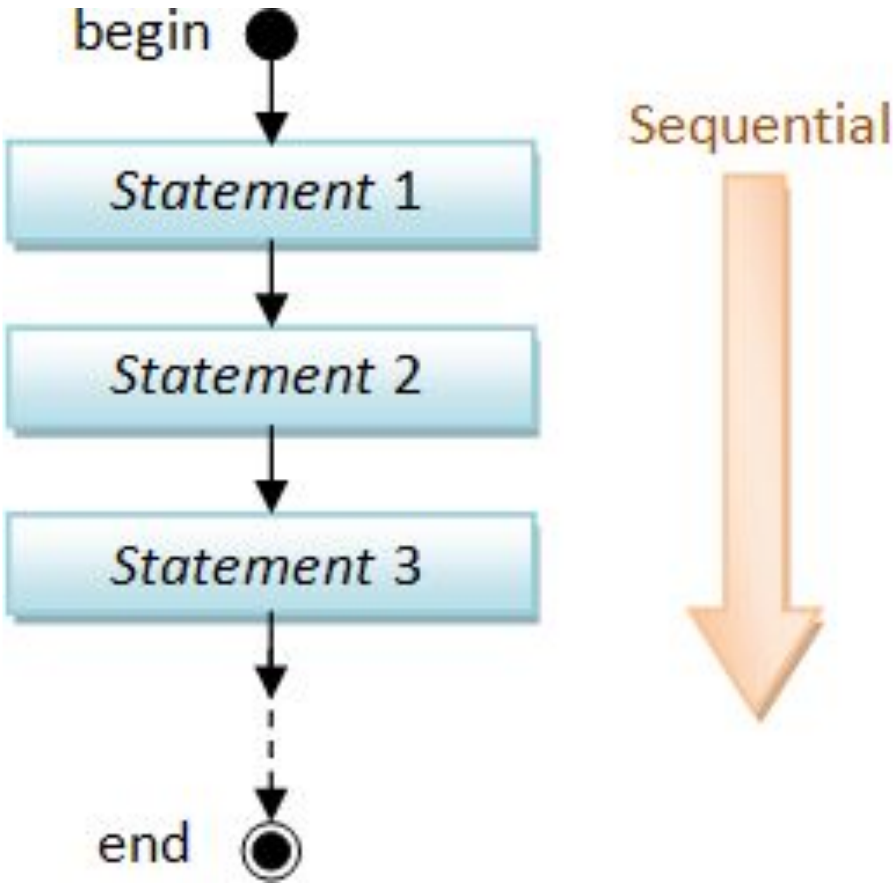
## Flowchart:



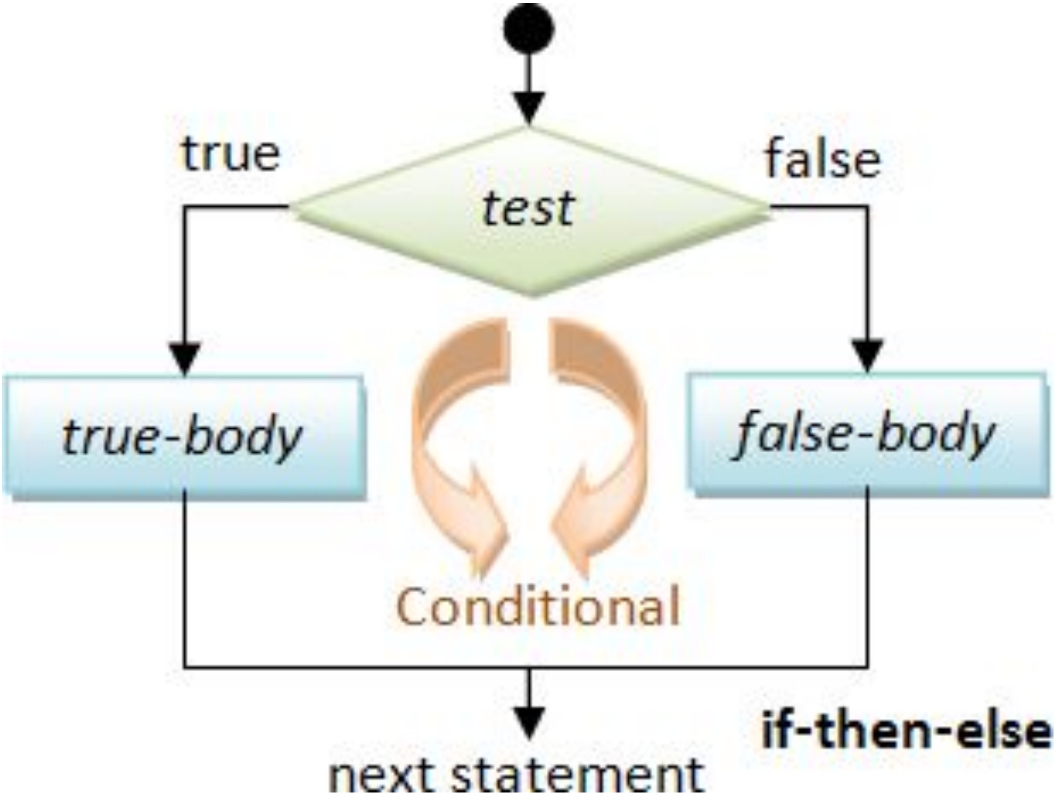

## Algorithm:

Step 1: Input M1,M2,M3,M4
Step 2: GRADE ← (M1+M2+M3+M4)/4
Step 3: if (GRADE <50) then
Print "FAIL"
else
Print "PASS"
endif

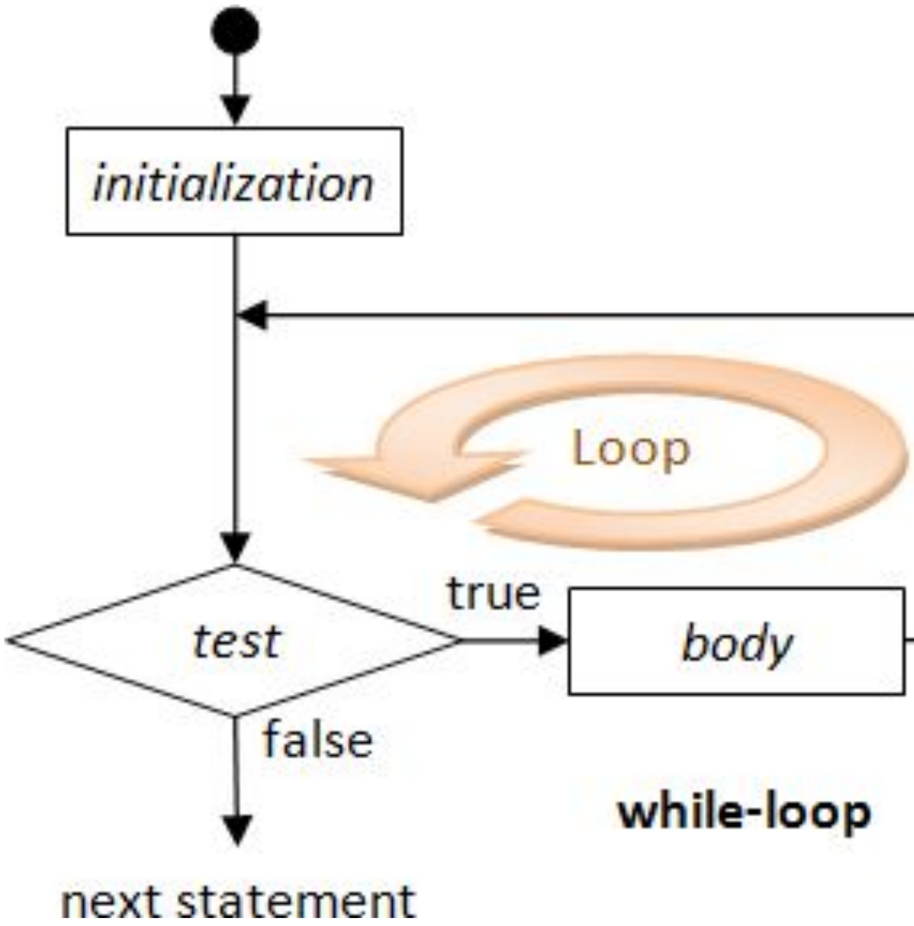
# SEQUENTIAL FLOW CHART



# CONDITIONAL FLOW CHART



# LOOPING FLOW CHART



# EXAMPLE

Write an algorithm anddraw a flowchart to convert the length in feet to centimeter.

Pseudocode:

Input the length in feet (Lft)

Calculate the length in cm (Lcm) by multiplying Lft with 30

Print length in cm (Lcm)

# EXAMPLE

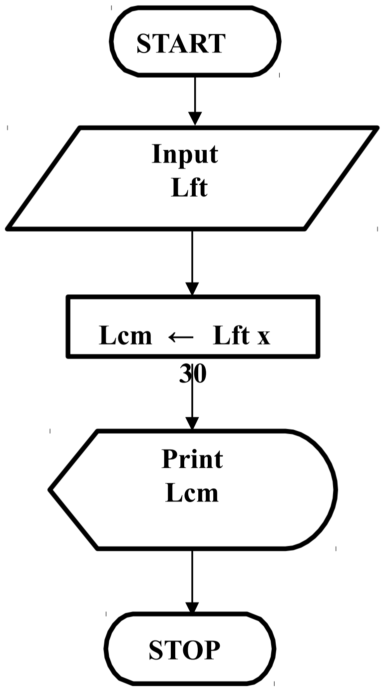
Algorithm

Step 1: Input Lft

Step 2: Lcm = Lft x 30

Step 3: Print Lcm

Flowchart



# WHY ALGORITHMS & FLOWCHARTS ?

A typical programming task can be divided into two phases:

- *Problem solving phase*
  - ✔ produce an ordered sequence of steps that describe solution of problem
  - ✔ this sequence of steps is called an *algorithm*
- *Implementation phase*
  - ✔ implement the program in some programming language

# Flowchart

1. Draw a flowchart to read a number and find the sum with its previous number.
2. Draw a flowchart to check if the given no. is positive or not?
3. Write an algorithm and draw a flowchart to find the biggest no from 2 given numbers.
4. Write an algorithm and draw a flowchart to find the sum of first N real numbers.
5. Write an algorithm and draw a flowchart to find the sum of first n odd numbers.
6. Write an algorithm and draw a flowchart to find the biggest among 3 numbers.

THANK YOU